

Andrew Ghobrial

Senior Backend Engineer | AWS Architect

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PROFESSIONAL SUMMARY

Senior Backend Engineer and AWS Architect with 7+ years building cloud-native systems. At Flowla, owns features end to end, from UI to infrastructure, including CRM integrations, AI content pipelines, SOC 2 recertification, and OpenTelemetry observability. Previously led a backend team across three products on serverless AWS with Node.js, TypeScript, Terraform, and CloudFormation, cutting non-production costs by 50%. Holds five AWS certifications, two at Professional level.

TECHNICAL SKILLS

Languages & Runtime: TypeScript, JavaScript, Node.js, Bun

Backend & APIs: NestJS, Express.js, REST APIs, Serverless Framework, Event-Driven Architecture, Socket.IO, Stripe, Salesforce, Attio, Slack API

AWS: Lambda, EC2, ECS, RDS, Aurora, DynamoDB, EventBridge, SNS, SQS, CloudFormation, Well-Architected Framework

Infrastructure & DevOps: Terraform, Infrastructure as Code (IaC), Docker, CI/CD, GitHub Actions, Multi-Region Deployment, OpenTelemetry (OTel), Observability

Databases: PostgreSQL, MongoDB, DynamoDB, Firebase

AI & Security: Vercel AI Gateway, Stable Diffusion, ComfyUI, SOC 2 Compliance

Also: GCP, React, React Native, Retool, Code Review, Technical Leadership

PROFESSIONAL EXPERIENCE

Senior Backend Engineer

Nov 2025 – Present

Flowla — London, United Kingdom (Remote)

- Own features end to end in direct contact with the product owner, building the UI alongside the backend services.
- Built AI content pipelines on Vercel AI Gateway that generate text and image content blocks tailored to each customer's needs.
- Developed core integration features, including a new Attio app integration and a new version of the Salesforce app.
- Closed security gaps across the platform to support the company's SOC 2 recertification.
- Revamped the infrastructure by refactoring Terraform templates around a proper module architecture, variable resolution, and state locking, and added GitHub Actions CI/CD pipelines for infrastructure deployments.
- Integrated OpenTelemetry (OTel) across backend services for end-to-end observability.
- Built a Slack bot that starts and stops infrastructure resources on automated schedules to reduce costs, with on-demand commands for starting and stopping environments.
- Developed learning management system (LMS) functionality within the product.
- Migrated the codebase from Yarn to Bun.

Backend Team Lead & AWS Architect

Aug 2023 – Oct 2025

DevGurus — United States (Remote)

- Led backend engineering across three products: GameSafe (child safety in online gaming), DCA (a digital citizenship learning platform), and GunSync (complex marketplace integrations).
- Reduced non-production AWS costs by 50% by optimizing EC2, ECS, and RDS resource utilization while maintaining application reliability.
- Migrated a legacy parental control application to a cloud-native, multi-region architecture provisioned with Terraform to meet compliance requirements.
- Architected applications on AWS following the Well-Architected Framework, implementing infrastructure with CloudFormation and the Serverless Framework.
- Led development of core backend APIs and conducted pull request reviews to maintain engineering standards across the team.
- Built CI/CD pipelines with GitHub Actions and improved application architecture for performance, scalability, and monitoring.

Senior Software Engineer

Nodogoro — Cairo, Egypt

- Led development of StudioShot, an AI product that generates professional headshots from selfies using Stable Diffusion and ComfyUI.
- Owned backend development and maintenance of AWS and GCP components, improving cost and performance by moving AI GPU workloads onto AWS EC2 GPU instances.
- Migrated the legacy application architecture to a scalable design built on AWS and GCP native services.
- Developed internal tooling with Retool to support operations and delivery teams.

May 2023 – Mar 2026

Backend Engineer (Freelance)

Upwork — Cairo, Egypt

- Delivered two products as a Top Rated freelancer: a marketplace app for home maintenance jobs and a mood-tracking app analyzing user mood against daily activity.
- Implemented backends in Node.js and TypeScript on AWS using the Serverless Framework and a cloud-native approach.

Jun 2021 – Aug 2022

Professional Services Delivery Engineer

Onica, a Rackspace Company — Cairo, Egypt

- Delivered two client engagements: a medical application integrating a Bluetooth-enabled inhaler, and a cost-simulation tool for underground water pipeline projects.
- Built event-driven, cloud-native architecture on AWS using the Serverless Framework with Lambda, DynamoDB, Aurora, EventBridge, SNS, and SQS for efficient event propagation.
- Developed frontends in React with TypeScript and backends in Node.js with TypeScript and MongoDB.

Sep 2020 – Apr 2023

Software Engineer

Callvita, LLC — Cairo, Egypt

- Led development of an application for storing patient medical history and booking doctor appointments.
- Built the web frontend in React, backend services in Node.js and Express, and the mobile app in React Native, hosted on AWS.

Jun 2019 – Aug 2020

AWS CERTIFICATIONS

AWS Certified DevOps Engineer – Professional	2024
AWS Certified Solutions Architect – Professional	2023
AWS Certified Developer – Associate	2023
AWS Certified Solutions Architect – Associate	2021
AWS Certified Cloud Practitioner	2021

BSc in Computer Science

German University in Cairo — Cairo, Egypt

Bachelor thesis in computer science and engineering

Sep 2014 – May 2019

Bachelor Project — Exchange Placement

DHBW Stuttgart — Stuttgart, Germany

Built a VR-based personal assistant for elderly users

Mar 2018 – Aug 2018

LANGUAGES

Arabic (Native) | English (Fluent)